PROJECT ENGINEERING & DOMAIN ANALYSIS REPORT

Automated Neural Image Colorization Pipeline via Deep Architecture

Project Type:	Computer Vision / Deep Learning	**Framework:**	PyTorch & Gradio Stack
Dataset:	Kaggle Landscape Pictures (4,000 Images)	**Hardware:**	NVIDIA Tesla T4 GPU (CUDA Accelerated)

1. EXECUTIVE SUMMARY

This technical report documents the engineering domains utilized in the design, training, validation, and deployment of an end-to-end automated image colorization system. The system maps single-channel grayscale lightness images to two-channel chrominance vectors using deep convolutional networks. By integrating advanced practices across five core engineering competencies, a high-fidelity **U-Net architecture** was trained over 50 epochs on 4,000 high-resolution landscape images, achieving an experimental Peak Signal-to-Noise Ratio ($PSNR$) of **28.26 dB** and a Structural Similarity Index ($SSIM$) of **0.3863**. This document categorizes the engineering implementations by domain to provide a granular review of the underlying codebase.

2. COMPUTER VISION & IMAGE PROCESSING DOMAIN

Image colorization is formulated not as a direct RGB pixel regression problem, but as a perceptually decoupled structural color mapping within a specialized color space.

- **Perceptual Color Space Decoupling:** The pipeline transforms input imagery from the standard RGB space to the **CIE LAB color space** using `skimage.color.rgb2lab`. The L channel represents luminance (lightness from 0 to 100) and is isolated as the structural grayscale input. The a (green-red spectrum) and b (blue-yellow spectrum) channels capture isolated chrominance information.
- **Channel Normalization Scaling:** To accommodate neural network convergence limits, the L channel is scaled to the interval `[0.0, 1.0]` via division by `100.0`, and the ab channels are scaled to the interval `[-1.0, 1.0]` via division by `128.0`. This ensures numerical uniformity and prevents gradient explosion.
- **Advanced Evaluation Metrics:** Traditional Mean Squared Error ($MSE = \Sigma(P - T)^2 / N$) is supplemented with strict structural and perceptual metrics from `skimage.metrics`:
  - **Peak Signal-to-Noise Ratio (PSNR):** Quantifies the logarithmic ratio between the maximum possible power of the signal and the corrupting noise ($PSNR = 10 \times log\&sub1;\&sub0;(MAX^2 / MSE)$), evaluated over a specific data range of `2.0`.
  - **Structural Similarity Index (SSIM):** Tracks luminosity, contrast, and structural degradation across local windows, ensuring structural coherence in the recolorized bounds.

> **Image Processing Insight:** Colorizing in the LAB space guarantees that the structural integrity of the original grayscale photo remains perfectly intact, as the network is mathematically restricted from modifying the lightness values.

3. DEEP LEARNING & NEURAL NETWORK ARCHITECTURE DOMAIN

The pipeline builds, tests, and validates complex neural networks to handle high-dimensional spatial translations using deep learning primitives.

- **Deep Architectures Implemented:** The codebase supports three architectures: a Baseline CNN, a traditional bottleneck Autoencoder, and an advanced **Colorization U-Net**.
- **U-Net Topology with Skip Connections:** The primary trained network (`ColorizationUNet`) utilizes an asymmetric contracting encoder path, an expansion decoder path, and direct **skip connections**. These connections concatenate early high-resolution spatial feature maps from the encoder with upsampled feature maps in the decoder. This architecture contains **1,861,762 parameters**, providing an optimal balance between spatial capacity and generalization.
- **Optimization & Multi-Stage Training:** The network minimizes a standard mean squared error criterion (`nn.MSELoss`) optimized via the `optim.Adam` optimizer initialized with a learning rate $\alpha = 0.005$. Regularization and convergence are accelerated by a learning rate schedule that decays weights uniformly over the 50-epoch budget.

4. DATA ENGINEERING & PIPELINE OPTIMIZATION DOMAIN

Data ingestion, dataset preparation, and streaming throughput were engineered to remove bottlenecks and maximize hardware efficiency.

- **Automated Data Ingestion:** Integration with the Kaggle API automated the secure authentication, forceful downloading, and unzipping of the 620MB `arnaud58/landscape-pictures` archive directly to the ephemeral environment storage disk.
- **Asynchronous Parallel Loaders:** Custom sub-classes of `torch.utils.data.Dataset` handle on-the-fly conversion of imagery from BGR to RGB via OpenCV, spatial resizing to a uniform `224×224` resolution, and extraction of LAB channels. These are fed to `DataLoader` objects operating with `num_workers=2` and `pin_memory=True`, enabling asynchronous CPU-to-GPU data transfers.
- **Strong Chrominance Augmentation:** To combat color convergence stagnation, a powerful geometric and color jitter augmentation pipeline was deployed using `torchvision.transforms`:
  RandomHorizontalFlip (p=0.5) **RandomRotation (15°)**
  ColorJitter (Brightness=0.2, Contrast=0.2, Saturation=0.3, Hue=0.05) **RandomAffine (Translate=10%)**
  The custom saturation jitter forces the network to learn rich, diverse color representations instead of predicting safe sepia tones.

5. DATA VALIDATION, QUALITY CONTROL & DIAGNOSTICS DOMAIN

Rigorous assertions, shape validations, and runtime health diagnostics protect the pipeline from structural faults or training collapse.

- **Pre-Training Structural Verification:** Prior to model optimization, the data loader pipeline is sanity-checked by extracting an isolated batch. It explicitly verifies that the input lightness tensor L conforms to `[64, 1, 224, 224]` and chrominance tensor ab conforms to `[64, 2, 224, 224]`, while validating that numerical distributions lie within expected scaled intervals.
- **Variance Decay Diagnostics:** The training loop monitors the moving standard deviation (**Prediction Std**) of output tensors. A drop in variance ($\sigma < 5.0$) automatically triggers system warnings to alert engineers to desaturated output, typical of structural underfitting or mode collapse.

6. FULL-STACK INTEGRATION & WEB UI DEPLOYMENT DOMAIN

The model was successfully transitioned from an experimental script into a complete, user-facing software artifact.

- **Interactive Gradio Interface:** A full-stack web UI was built using `gradio`, exposing a clean drag-and-drop panel for users to upload arbitrary grayscale images and receive real-time colorized conversions.
- **Robust Serialization & Logging:** The framework automatically logs loss metrics, learning rates, execution durations, and prediction variances into structured JSON files. It implements an automated model checkpointing system, archiving both the final iteration weights and the state dict corresponding to the absolute minimum validation loss (`UNet_best.pth`).

7. EXPERIMENTAL RESULTS & PERFORMANCE MATRIX

The table below outlines the compiled performance metrics extracted directly from the system logs following the completion of the 50-epoch training schedule.

Model Architecture	Parameters	Best Val Loss	Final Val Loss	Best PSNR	Final PSNR	Best SSIM	Final SSIM	Total Time
UNet (Skip Connections)	1,861,762	0.0443	0.0443	28.26 dB	28.21 dB	0.3863	0.3830	20.43 min

> **Key Success Metric:** The U-Net architecture emerged as the dominant model, achieving the lowest loss, highest peak signal-to-noise ratio, and maximum structural similarity index due to its capacity to retain high-frequency spatial details from skip connections.